Energy Decay in a Damped Harmonic Oscillator

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Abstract

We study the free motion of a mass on a linear spring subject to viscous damping. We derive the closed-form solution in the underdamped regime, show that the mechanical energy decays monotonically, and compare the analytic solution with two simple numerical schemes.

1 Introduction

The damped harmonic oscillator is the simplest model of a system that loses energy to its surroundings [1]. Its equation of motion is

$$m\ddot{x} + c\dot{x} + kx = 0, \tag{1}$$

where $m > 0$ is the mass, $c \geq 0$ the damping coefficient and $k > 0$ the spring constant.[1] Dividing by m and writing $\omega_0 = \sqrt{k/m}$ and $\gamma = c/(2m)$ gives the *standard form*

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0. \tag{2}$$

2 Solution

Substituting $x = e^{\lambda t}$ into (2) gives the characteristic equation $\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$, whose roots are

$$\lambda_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{3}$$

The behaviour depends on the sign of the discriminant:

$$regime = \begin{cases} \gamma < \omega_0, \\ \gamma = \omega_0, \end{cases}$$

In the underdamped case let $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$. With $x(0) = x_0$ and $\dot{x}(0) = 0$ the solution is

$$x(t) = x_0 e^{-\gamma t}(\cos)\omega_d t + \frac{\gamma}{\omega_d \sin}\omega_d t. \tag{4}$$

Equivalently, with $y = (x, \dot{x})^{T}$, equation (2) becomes the first-order system

$$\dot{y} = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 - 2\gamma & \end{pmatrix} y. \tag{5}$$

Theorem 1. *The energy $E(t) = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2$ never increases.*

[1] All quantities are in SI units.

References

[1] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics, 3rd ed.*, Addison-Wesley, 2002.

[2] E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I, 2nd ed.*, Springer, 1993.